38N

ARLON ELECTRONIC MATERIALS

Polyimide Low-Flow Prepreg

38N is a polyimide low-flow prepreg suitable for bonding multilayer polyimide rigid-flex, attaching heat sinks to polyimide MLBs, or other applications where minimal and uniform resin flow is required.

Features:

- Novel chemistry ensuring faster and more uniform resin cure for minimal and consistent resin flow, preventing excessive flow into clearance/relief areas
- Tg = 200°C and expansion characteristics typical of polyimide greatly improves PTH reliability
- Curable at temperatures as low as 350°F (177°C)
- Excellent thermal stability
- Improved bond strength to copper and other metals for excellent performance in heat sink bonding applications
- Electrical and mechanical properties meeting the requirements of IPC-4101/42
- Compatible with lead-free processing
- RoHS/WEEE compliant

Typical Applications:

- Bonding multilayer polyimide rigid-flex
- Finished PCB assemblies requiring excellent thermal stability
- Dielectric insulators
- Other applications where minimal and uniform resin flow is required

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Typical Properties:

Property	Units	Value	Test Method
Electrical Properties			
Dielectric Constant @ 1 MHz	-	4.25	IPC TM-650 2.5.5.3
Dissipation Factor @ 1 MHz		0.01	IPC TM-650 2.5.5.3
Volume Resistivity			
C96/35/90	MΩ-cm	8.2×10^7	IPC TM-650 2.5.17.1
E24/125	MΩ-cm	4.7×10^9	IPC TM-650 2.5.17.1
Surface Resistivity			
C96/35/90	MΩ	4.4×10^6	IPC TM-650 2.5.17.1
E24/125	MΩ	1.2×10^9	IPC TM-650 2.5.17.1
Electrical Strength	Volts/mil (kV/mm)	1600 (63.0)	IPC TM-650 2.5.6.2
Arc Resistance	sec	125	IPC TM-650 2.5.1
Thermal Properties			
Glass Transition Temperature (Tg)			
TMA	°C	200	IPC TM-650 2.4.24C
Decomposition Temperature			
Initial	°C	311	IPC TM-650 2.4.24.6
5% weight loss	°C	330	IPC TM-650 2.4.24.6
T260	min	50	IPC TM-650 2.4.24.1
T288	min	5	IPC TM-650 2.4.24.1
T300	min	3	IPC TM-650 2.4.24.1
CTE (X,Y)	ppm/°C	17	IPC TM-650 2.4.41
CTE (Z)			
< Tg	ppm/°C	54	IPC TM-650 2.4.24C
> Tg	ppm/°C	157	IPC TM-650 2.4.24C
z-axis Expansion (50-260ºC)	%	1.5	IPC TM-650 2.4.24C
Mechanical Properties			
Peel Strength to Copper (1 oz/35 micron)			
After Thermal Stress	lb./in (N/mm)	8.5 (1.5)	IPC TM-650 2.4.8C
Peel Strength to Kapton			
As Received	lb./in (N/mm)	5.9 (1.0)	IPC TM-650 2.4.9E
After Solder	lb./in (N/mm)	5.2 (0.9)	IPC TM-650 2.4.9E
Young's Modulus CD/MD	Mpsi (GPa)	2.1 (14.5)	ASTM E111
Flexural Strength	kpsi (MPa)	60 (414)	ASTM D3039
Tensile Strength CD/MD	kpsi (MPa)	32 (221)	ASTM D3039
Poisson's Ratio	-	0.18	ASTM E13204
Physical Properties			
Water Absorption (0.062")	%	<1.0	IPC TM-650 2.6.2.1A
Density	g/cm^3	1.6	ASTM D792 Method A
Thermal Conductivity	W/mK	0.3	ASTM E1461
Flammability	class	Meets V0	UL-94

Results listed above are typical properties, provided without warranty, expressed or implied, and without liability. Properties may vary, depending on design and application. Arlon reserves the right to change or update these values.

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Availability:

Arlon Part Number	Glass Style	Resin (%)	Pressed Thickness	Flow Range
38N0666	106	66	1.9	70 - 120 mils
38N8060	1080	60	2.9	70 - 120 mils

Recommended Process Conditions:

Because of varying storage conditions, it is recommended that 38N prepreg be dried at 29” (736mm) Hg for 12 to 24 hours.

38N Low-Flow prepreg is very process tolerant. It laminates well with either a cold platen press start or with a hot start. Vacuum or vacuum assist lamination is recommended for the removal of moisture and air. Low-Flow products do not displace air voids as well as standard prepregs, and vacuum will help assure a void-free final product.

Lamination Cycle:

1) Vacuum draw down the package for 30 minutes at <29” (736 mm Hg) prior to applying pressure in the press. Maintain the vacuum beyond the set point of the resin, i.e., above 320°F (160°C)
2) Use a platen temperature in the range of 370°F - 380°F (188°C - 193°C).
3) Control the heat rise to about 8°F - 12°F per minute (4.5°C - 6.5°C) between 210°F and 300°F (100°C and 150°C)
4) Use a pressure of 180 to 350 psi (12.6 to 24 kg/sq.cm), depending on panel size and complexity. Following are recommended pressures relative to panel size to use as starting points:

Panel Size		Pressure	
in.	mm	psi	kg/cm2
12 x 18	305 x 457	250	18
16 x 18	406 x 457	290	20
18 x 24	457 x 610	330	23

Cure time is 90 minutes at temperature.

The subsequent processing should be the same as those normally used for rigid-flex PCBs